

 सत्यमेव जयते	भारत सरकार दूरसंचार विभाग दूरसंचार अभियांत्रिकी केंद्र Government of India Department of Telecommunications Telecommunication Engineering Centre K.L. Bhawan, Janpath, New Delhi- 110001	 ISO 9001:2015
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No. 26-1/2022-FA/TEC

Date: 28.03.2024

Subject: Minutes of Ninth (9th) meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group (SG)-15 held on 27.03.2024.

The Ninth meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group 15 regarding “*Networks, Technologies, and Infrastructures for Transport, Access & home*”, reconstituted for the Study Period 2022-24, was held on 27.03.2024 under the chairship of Sh. Rakesh Desai, DDG(FA), TEC through video-conferencing. The list of participants is placed at Annexure A.

2. At the outset, Sh. Rajesh Tripathi, Director (FA) TEC & Convener of NWG-15, welcomed all the participants and briefly explained about the agenda and further proceedings of the meeting.

3. Sh. Rakesh Desai, Chair/ NWG-15 & DDG (FA) TEC, greeted all the members and welcomed all participants especially new member Mr. Anil Gehlot, DDG (CS) DOT HQ. He briefly informed achievements of NWG15 in submitting contributions at ITU-T SG15 leading to be part of the final recommendations or the final recommendation of ITU-T SG15. Sh. Desai, in his opening remarks, highlighted that two contributions have been submitted by India in last (3rd) meeting of ITU-T SG15 held at Geneva and actively participated in Q7/15 discussions for another two contributions and revision of texts of different questions. Sh. Desai also presented the executive summary of 3rd meeting of ITU-T SG15. He apprised the participants that 13 delegates from India have attended 3rd meeting of ITU-T SG15 either in-person or virtually. He further informed that the next meeting of ITU-T SG15 is scheduled at Montreal, Canada from 1-12 July 2024. He also requested the members to attend online correspondence discussion meetings called by the concerned Rapporteurs between April to June 2024.

After the introduction, Sh. Rakesh Desai explained the members on potential work-items for contributions and also requested other members to actively participate in the discussions on various contributions.

4. Contribution on- ‘L.nis-Technical and functional requirements for network infrastructures sharing’ (Q7/15)

Sh. Desai explained that Indian contributors have furnished responses to the Questionnaire circulated by TSB in consultation with CS Division of DOT HQ, TSPs / ISPs & NWG-15 members as discussed in 8th meeting of NWG-15 dated 19.10.2023 and as an outcome to the discussions in the 3rd meeting of SG15, a new work-item – ‘L.nis-Technical and functional requirements for network infrastructures sharing’ has been initiated in Q/15. India is also one of the supporting members of this new work-item. As per the A.1 justification of this new work-item, the probable shared network infrastructures may contain tower, pole,

passive component, optical fibre cable, duct and so on. Participants were informed that India would furnish inputs / contribution on the technical and functional requirements for the network infrastructures that could be shared among multi operators (TSPs / ISPs/ IPs).

Members of NWG-15 were requested to come forward and furnish their inputs for this work-items. The contributors are requested to prepare a draft contribution **by 20th April 2024** and participate in correspondence discussions on this work-item.

(Action by: Sh.Rakesh Desai, Sh. Anil Gehlot, Ms. Mudita Chandra, any other interested NWG-15 members)

5. Correspondence activity of ‘G.Suppl.G65x: Standardization roadmap for SDM optical fibres including Multi-Core Fibres ‘ (Q5/15)

Sh. Desai discussed the draft working document circulated by Q5/15 wherein new supplement G Suppl.G.65x on ‘Standardization roadmap for optical fibres and cables for space division multiplexing’ has been proposed as per the decision of 3rd meeting of SG15. Sh.Desai informed that Prof. Deepa from IIT-Madras and Mr.Arvind from STL has pointed out for possible contribution on Multi-Core Fibres (MCF) as such fibre is being studied in DOT’s ongoing project namely ‘Advance Optical Communication Test Bed Project. Initial inputs received from Mr. Sobhanbabu, IIT-Madras were briefly mentioned during discussions.

Sh.Desai requested Mr Bhaumik from M/s STL to request Mr Arvind of M/s STL on behalf of NWG-15 to contribute in this work-item or nominate one expert from STL in this regard. He further requested the interested members to discuss this work-item among small group in focused manner so that quality inputs may be furnished during correspondence discussions and draft contribution can be prepared by **30th April 2024**. Members of NWG-15 were requested to come forward and furnish their inputs for this work-items.

(Action by: Sh.Rakesh Desai, Prof. Deepa / Mr. Sobhanbabu / experts from IIT-Madras, Sh.Arvind / Sh. Sudipta Bhaumik from STL, any other interested NWG-15 members)

6. Contribution on- ‘G.fso-Terrestrial free space optics for mobile backhaul with short reach interfaces’ (Q6/15)

Sh.Desai informed the participants that editors of new work-item – G.fso (Q6/15) sought inputs / suggestions from Indian experts on optical interface specifications and requirements for free space optics (FSO) short-reach terrestrial applications. Sh.Desai requested experts from Industries (M/s Navtech, M/s Velminni), Researchers (IIT-D, IIITD, ERNET, etc) to come forward, discuss in small group, participate during correspondence discussions and prepare draft contribution based on the discussions / deliberations in the Q6/15 meeting scheduled in the last week of April 2024.

(Action by: Sh.Rakesh Desai, Dr. A. Paventhan from ERNET, Experts from Industries (M/s Navtech, M/s Velminni), Researchers from IIT-D / IIITD, any other interested NWG-15 members)

7. Contribution on - Revision of Recommendation L.360: “Operations support system requirements for infrastructure and network elements management using Identification data (ID) technology” (Q7/15)

Sh. Desai apprised the participants regarding discussions in Q/15 meeting during 3rd meeting of SG15 in respect of Indian inputs on the revision of Recommendation L.360:

“Operations support system requirements for infrastructure and network elements management using ID technology”. He stated that Q7/15 requested Indian delegates to submit a fresh contribution clarifying how the ID tags work in the proposed operation system and investigate the relationship with responsibility of Q2/15. Sh.Desai also presented the initial draft of new contribution on ‘Network Monitoring Expert (NME)’ and requested contributors to improve upon draft further by 30th April 2024.

(Action by: Sh.Rakesh Desai, TEC and S/Sh. Guharajan Shivakumar & Sriram from M/s Aprecomm.)

8. Contribution on ‘Revision of Recommendation for L.105 Amd.1. Optical fibre cables for drop applications’ and revisions of other L-Series recommendations (Q5/15)

Sh. Desai apprised the members that last Q5/15 meeting during 3rd meeting of ITU-T SG15 proposed a revision plan for L.104/L.67, L.105/L.87, L.107/L.78, L.108/L.79, and L.111 to maintain the user convenience for these Recommendations. Sh.Desai specifically discussed A.1 justification for revision of Recommendation for L.105 Amd.1. Optical fibre cables for drop applications and highlighted that L.105/L.87 describes the characteristics, construction and test methods of optical fibre cables for drop applications.

Members of NWG-15 were requested to come forward, participate during correspondence discussions and prepare draft contribution for revisions of the above-stated L-series recommendations based on the discussions / deliberations in the Q5/15 meeting scheduled in the last week of April 2024.

(Action by: TEC: Sh.Rakesh Desai, Ms. Mudita Chandra, Experts from Tx Division; Sh. Sudipta Bhaumik from STL, experts from HFCL, IIT-Madras, any other interested NWG-15 members)

9. Contribution on proposed new supplement- ‘G.sup.PON security- Practical aspects of PON security’ (Q2/15)

Sh. Desai apprised the members that last Q2/15 meeting during 3rd meeting of ITU-T SG15 proposed a new supplement- ‘G.sup.PON security- Practical aspects of PON (passive optical network) security’ and also revision of existing G-series recommendations viz. G.984.3, G.987.3, G.9807.1, and G.988. Sh.Desai presented the draft document circulated by the editors of Q2/15 stating further that the proposed new supplement envisaged to address the system invariant aspects of the threat model in PON, the operator expectations about the security of the PON-based optical access networks, the method how to utilize the techniques to achieve the best possible security performance, feasible innovative technical extensions/mechanisms which may be considered in future PON recommendations to support security enhancement features.

Sh.Desai requested experts from CDOT and NCCS (National Centre for Communication Security) to come forward and furnish their inputs / contributions in this proposed new supplement on PON security.

(Action by: Sh.Rakesh Desai, TEC, Sh. Guharajan Shivakumar from M/s Aprecomm, Experts from CDOT and NCCS)

10. Contribution on revision of Technical Paper ‘LSTP-GLSR- Guide on the use of ITU-T L-series Recommendations related to optical technologies for outside plant’ (Q5,7,8/15)

Sh. Desai discussed the draft working document circulated by the editors of LSTP-GLSR (Q5/15) stating further that the said technical paper provides information on the background, development and uses of L series Recommendations.

Members of NWG-15 were requested to study the draft technical paper and come forward for contributing in this work-item.

(Action by: All members)

11. Contribution on distributed fibre optic sensing (DFOS) systems (Q6/15)

Sh.Desai informed the members that Indian contribution on DFOS was well-received in the 3rd meeting of ITU-T SG15. He also highlighted that several contributions were submitted by different countries on DFOS that were discussed in Q2,5,6,7,8,12 & 14 meetings besides joint meetings. He stated that SG15 agreed to revise the ToR of relevant questions of SG15 adding distributed fibre optic sensing technology for next study period and also agreed to start a new work item on DFOS in Q6/15 viz. G.dfos-Distributed fibre optic sensing system for terrestrial optical transmission system, covering DFOS optical interface requirements for implementing DFOS within terrestrial optical transmission systems.

Sh.Desai requested the contributors to contribute in the new work-item G.dfos during future consultations / discussions of Q6/15 meetings.

(Action by: Contributors: Experts from M/s Sterlite Technologies Limited (STL) viz. S/Sh. Badri Gomatam; Nagaraju Bezawada; Abhishek Sati; Ashit Padhi; Sudipta Bhaumik and Sh.Rakesh Desai from TEC)

12. Contributions to APT for the forthcoming WTSA-24:-

Sh. Desai, while referring to earlier discussions, pointed out that India is hosting next WTSA meeting scheduled to be held in Oct 2024 at New Delhi. He presented drafts of two WTSA resolutions viz. Resolution no.40- Regulatory & policy aspects of the work of the ITU-T and Resolution no.68- Evolving role of industry in the ITU-T and informed that these drafts were prepared based on the inputs from experts of NWG-15 and presented virtually during APT meeting held on 5-6th February 2024. He requested all NWG-15 members to examine existing resolutions of WTSA and submit inputs/ suggestions on WTSA resolutions related to SG15 by 15th April 2024 in view of forthcoming APT meeting scheduled in the last week of April 2024.

(Action by: All members)

13. Sh. Rakesh Desai requested all the NWG-15 members to obtain TIES account by registering on ITU portal and also register for the forthcoming WTSA-24 meeting. He also urged all NWG-15 experts to participate actively in group activities and submit more contributions. He once again requested NWG-15 members to register on 'Standards Coordination Portal' of TEC (<https://tec.gov.in/scp/>) and start familiarizing with the features of the portal.

14. The key action points emerged from the meeting are as under: -

- i. Members / experts are requested to contribute on the work-items as per the timeline as discussed at para 4 to 11 above.
- ii. Members may examine the WTSA resolutions and propose amendments or new resolutions pertaining to SG15.

- iii. NWG-15 Members may please register on 'Standards Coordination Portal' of TEC (<https://tec.gov.in/scp/>) and start familiarizing with the features of the portal.
 - iv. Members may study all the questions of ITU-T SG15 and come forward to increase number of Indian contributions in ITU-T SG15.
15. The meeting ended with the vote of thanks to the chair.



S/d

(Rajesh Tripathi)

Convener, NWG-15 & Director (FA), TEC

To,

All the participants / Members of the NWG-15.

Copy to: Sr.DDG & Head, TEC for kind information.